Md Ahadul Haque

Software Engineer

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Professional Summary

Software Engineer with 3+ years of experience in AI engineering, full-stack development, and data engineering. Designs and builds agentic AI systems and LLM-powered applications, including multi-step agent workflows, retrieval-augmented generation (RAG), and tool-integrated assistants, with AI features shipped to production. Develops scalable full-stack web applications end to end, from backend APIs and databases to responsive frontends and cloud deployment. Experienced in building ETL pipelines and orchestrating data workflows for large-scale data processing and analytics.

Skills

- **Languages:** TypeScript, JavaScript, Python, C/C++
- **Backend:** Node.js, NestJS, Express.js, FastAPI, Flask, Django, Microservices Architecture
- **APIs:** REST, GraphQL
- **Frontend:** React.js, Next.js, Tailwind CSS, Ant Design, Redux Toolkit, Zustand, Web Accessibility (WCAG 2.1)
- **Databases & ORMs:** PostgreSQL, MySQL, SQLite, MongoDB, Redis, Elasticsearch; Prisma, Sequelize, SQLAlchemy, Mongoose
- **AI/LLM:** LLM Integration (Cloud & Local Models), Prompt Engineering, RAG, LangChain, LangGraph, MCP, Vector Databases (Pinecone, pgvector), Claude Code (Agentic Coding), Ollama
- **MLOps:** Model Serving (FastAPI + Uvicorn, KServe), Kubeflow Pipelines, Model Versioning & Lifecycle Management, CI/CD for ML
- **Data Engineering & Automation:** Apache Airflow, ETL Pipelines, Web Scraping, Pandas, BeautifulSoup4, lxml, n8n
- **DevOps & Cloud:** Docker, Kubernetes, AWS, NGINX, CI/CD, Vercel, Netlify, Coolify, VPS (Hostinger, DigitalOcean)
- **Monitoring & Observability:** Prometheus, Grafana, Loki, Promtail
- **Auth & Security:** JWT, OAuth 2.0, Passport.js, NextAuth.js, ClerkJS, RBAC/ABAC, OWASP Top 10
- **Testing:** Jest, Pytest, Cypress, Playwright

Experience

Software Engineer

March 2026 – Present

Cefalo Bangladesh LTD

Tools: TypeScript, NestJS, Prisma (MySQL), Next.js, Ant Design, TanStack Query, Zustand, AWS S3, Docker, Turborepo, JWT, Google Calendar API, ElevenLabs

- **Recruitment Portal (Applicant Tracking System):** Contributed as a team member to a large-scale, production-grade ATS featuring a public job portal and an internal HR/admin dashboard, supporting end-to-end recruitment workflows.
- Developed and extended RESTful APIs across multiple NestJS modules, covering areas including candidate management, job postings, interview scheduling, evaluations, RBAC, and ABAC.
- Worked on the integration of Google Calendar API for automated interview scheduling and HackerRank API for in-platform coding assessments.
- Implemented features for multi-stage candidate tracking, dynamic application forms, and bulk Excel/PDF export for HR/admin reporting.
- Developed an AI-powered candidate shortlisting and scoring feature that automatically evaluates applicants against job requirements, reducing manual screening effort for the HR team.
- Integrated ElevenLabs to build an AI-powered voice interview feature that conducts automated screening interviews with candidates, enabling HR to assess applicants at scale without scheduling live interviewers for early-stage rounds.

Data Engineer

August 2024 – February 2026

Volue ASA, Norway (via Cefalo Bangladesh LTD)

Tools: Python, BeautifulSoup4, Pandas, Apache Airflow, Django, REST APIs, Docker, Grafana, Prometheus, GCP

- Collected, preprocessed, and structured high-resolution power market data from diverse sources into standardized formats.
- Designed, maintained, and enhanced ETL pipelines to support forecasting models and energy market analytics.
- Implemented new features within existing data workflows to improve reliability, scalability, and performance.
- Ensured data integrity and consistency across high-volume transformation processes.
- Orchestrated and scheduled complex data workflows using Apache Airflow across thousands of data sources, improving pipeline observability and reducing manual intervention.
- Containerized data services with Docker for consistent deployment across development and production environments on Google Cloud Platform (GCP).
- Set up monitoring and alerting with Prometheus and Grafana to track pipeline health and detect data anomalies in real time.
- Collaborated with cross-functional teams to transform energy market forecasting requirements into robust data engineering solutions.

Associate Software Engineer October 2023 – July 2024

Cefalo Bangladesh LTD

- **Team Task Summarizer (AI-Driven Automation):** Developed a secure automation system to aggregate and summarize team updates from GitHub, Slack, and ClickUp using OAuth 2.0 integrations. Built a data processing pipeline leveraging local LLMs (Llama 3.2 and Gemma) with structured prompt engineering to generate high-accuracy summaries. Delivered consolidated AI-generated reports to Gmail with zero external API cost through secure local model execution.
- **Automated Catalog Management:** Integrated ServiceNow with external service providers to automate request workflows, user data population, and approval processes, significantly reducing manual intervention.
- **Cefalo Movie Snap:** Engineered a scalable scraping system to extract Academy Award data from Wikipedia and enrich it via REST APIs; containerized the solution using Docker for deployment efficiency.
- **Blogger Platform:** Developed a responsive full-stack web application with JWT-based authentication and dynamic data sorting using React.js and Node.js.

Projects

AI Teaching Assistant — Agentic RAG Study Platform GitHub 2026

Python | FastAPI | LangGraph | LangChain | RAG | Gemini | OpenRouter | Pinecone | MongoDB Atlas | Streamlit | Pytest

- Built an AI study platform that turns static course material (PDF/DOCX) into an interactive tutor, letting students get cited, page-level answers instead of manually searching long documents, and test themselves with auto-graded quizzes; runs entirely on free tiers.
- Designed a Corrective RAG workflow in LangGraph that reduces hallucinated answers by grading retrieved chunks with an LLM, rephrasing weak queries, and falling back to web search when the uploaded documents lack the answer.
- Engineered an async ingestion pipeline that extracts text from PDF/DOCX files, detects scanned PDFs, and stores Gemini embeddings in per-document Pinecone namespaces to keep each user's data isolated, with chat history and quizzes persisted in MongoDB Atlas.
- Developed a provider-agnostic LLM layer with automatic Gemini → OpenRouter fallback and retry logic to keep the platform available under free-tier rate limits; secured it with ownership-checked document access, prompt-injection protection for web results, and per-user upload quotas.

Shikkhakosh — Full-Stack E-Commerce Platform Live Site 2026

Next.js 14 | NestJS | TypeScript | Prisma | MySQL | Tailwind CSS | Docker | Turborepo | Coolify | Hostinger

- Co-founded and architected a production-deployed e-commerce monorepo with a customer storefront, admin dashboard, and REST API; structured using Turborepo with shared UI component and TypeScript type libraries across all apps.
- Engineered a NestJS backend with 20+ modules covering the full commerce lifecycle — including product variants, hierarchical categories, coupon/reward systems, review moderation, push notifications, and custom invoice generation.
- Integrated Steadfast Courier API for bulk order dispatch and cancellation directly from the admin panel, and implemented JWT authentication with refresh token rotation and Google/Facebook OAuth support via Passport.js.
- Containerized the entire stack with Docker and deployed to a Hostinger VPS via Coolify, maintaining separate development and production environments at shikkhakosh.com.

Education

🎓 **B.Sc (Engineering) in Information and Communication Technology (ICT)** 2018 – 2023

Mawlana Bhashani Science and Technology University, Tangail-1902

Certifications

- **LLM Observability and Cost Management: Langfuse, Monitoring** – Udemy | Date: 09/11/2026 | Link
- **MLOps – Zero to Hero** – Udemy | Date: 25/10/2026 | Link
- **LangChain - Agentic AI Engineering with LangChain & LangGraph** – Udemy | Date: 03/10/2026 | Link
- **FastAPI with GenAI and AgenticAI Project – From Basic to AI** – Udemy | Date: 26/09/2026 | Link
- **Claude Certified Architect - Foundations** – Anthropic | Date: 07/09/2026 | Link
- **NestJS Mastery: Build & Deploy a Production-Ready API** – Udemy | Date: 16/03/2026 | Link
- **Claude Code In Action** – Anthropic | Date: 01/03/2026 | Link
- **Intermediate Secure Coding in NodeJS** – Secureflag | Date: 07/02/2026 | Link
- **Complete React, Next.js & TypeScript Projects** – Udemy | Date: 09/11/2025 | Link
- **Python Django – The Practical Guide** – Udemy | Date: 30/06/2025 | Link
- **OWASP Top 10:2021 in Python Learning Path** – Secureflag | Date: 10/09/2024 | Link